Roll No. | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.-6)

PHARMACEUTICAL BIOTECHNOLOGY-THEORY

Subject Code : BP605T

M.Code : 77990

Date of Examination : 23-05-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Explain the following terms in brief:**

   a. Biotechnology

   b. Transformation

   c. Immunological tolerance

   d. Monoclonal antibody

   e. Biosensors

   f. Plasma substitute

   g. rDNA

   Distinguish between :

   h. Antigen and antibody

   i. Toxin and Toxoid

   j. Vaccine and enzymes

SECTION - B

2. Explain the isolation of Glutamic acid by fermentation technology.

3. Enumerate how recombinant DNA technology used to produce Insulin.

4. **Write short note on:**

   a. Amylase

   b. Vitamine B12

SECTION - C

5. Explain the use of culture system for the production of microbial products.

6. How genetic recombination is used to prepare hepatitis B?

7. Enumerate role of steroids in the biotransformation processes.

8. Highlight the techniques of immobilization of enzymes.

9. What is PCR? Give its applications.

10. Explain structure and function of MHC.

11. Write short note on genetic engineering.

12. Give microbial production of peroxidase.

13. **Write a short note on :**

    a) ELISA

    b) Dried human plasma

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

e-2025